FIXTURE EVALUATION / 26 OCT 2026

A small deterministic baseline, measured honestly

The evaluation checks whether the implementation follows an explicit, reproducible synthetic label set. It does not estimate real-world accuracy.

Dataset and measured result

80 fictional assets x 3 real public CISA advisory fixtures = 240 in-process comparison pairs. The labeled set has 14 asset/advisory cases: 4 applicable, 4 not applicable, 6 manual verification.

Measure	Result	Scope
Labeled classification accuracy	100%	14 authored synthetic cases
Applicable precision / recall	100% / 100%	4 positive examples
Applicable false-negative rate	0%	In this fixture only
Expected uncertainty abstention	6/6	Missing/stale/ambiguous cases
Source traceability	14/14	ID + URL + excerpt present
Matcher benchmark	1.01 ms total	One 240-pair in-process pass

Interpretation: 14/14 is evidence that this fixed test set passes, not that OTProof will achieve 100% precision or recall on operator inventories. Four positive labels are far too few for a field performance claim.

AI extraction

Local Ollama inference was not run. No advisory extraction accuracy, summary quality, or AI productivity benefit is claimed. Deterministic CISA HTML/CSAF parsing is covered by unit tests, but no independent multi-vendor extraction gold set has been labeled.

CASE REGISTER

All labeled examples and confusion counts

The examples exercise exact affected product/version, outside-bound versions, a curated spelling alias, a similar but different product, missing/unsupported details, unknown model, stale inventory, and duplicate assets.

Asset	Advisory	Expected	Observed	Source linked
CV-HMI-001	ICSA-24-226-06	applicable	applicable	Yes
CV-HMI-002	ICSA-24-226-06	not applicable	not applicable	Yes
CV-HMI-003	ICSA-24-226-06	applicable	applicable	Yes
CV-HMI-004	ICSA-24-226-06	not applicable	not applicable	Yes
CV-HMI-005	ICSA-24-226-06	requires manual verification	requires manual verification	Yes
CV-HMI-006	ICSA-24-226-06	requires manual verification	requires manual verification	Yes
CV-HMI-007	ICSA-24-226-06	requires manual verification	requires manual verification	Yes
CV-HMI-008	ICSA-24-226-06	requires manual verification	requires manual verification	Yes
CV-HMI-009	ICSA-24-226-06	requires manual verification	requires manual verification	Yes
CV-OPT-010	ICSA-24-256-24	applicable	applicable	Yes
CV-OPT-011	ICSA-24-256-24	not applicable	not applicable	Yes
CV-APP-012	ICSA-24-284-15	applicable	applicable	Yes
CV-APP-013	ICSA-24-284-15	not applicable	not applicable	Yes
CV-OPT-014	ICSA-24-256-24	requires manual verification	requires manual verification	Yes

Three-class confusion matrix

Expected \ Observed	Applicable	Not applicable	Manual review
applicable	4	0	0
not applicable	0	4	0
requires manual verification	0	0	6

No semantic similarity model was used. Manual review is an intentional output for uncertain inventory and configuration cases.

Security and availability checks

The test suite checks local software behavior and known edge cases. It does not certify the system or validate real OT outcomes.

Checks included

Prompt-injection-like instructions are retained as untrusted advisory content; the extraction adapter rejects unsupported source values and quotations. Duplicate inventory IDs and macro-enabled workbooks are rejected. A failed source refresh serves a persisted stale snapshot. Role tests cover unauthorized remediation approval/verification, and the workflow API test records a human applicability decision, ticket approval, simulated evidence, and approver verification.

Performance scope

1.008 ms elapsed for one in-process pass over 240 pairs (0.0042 ms/pair on this developer machine). The measurement excludes database queries, HTTP ingestion, browser rendering, concurrency, cold starts, and human review. Operator review time was not instrumented or measured.

Limits

Small synthetic sample; three Rockwell CISA fixtures; no blinded ground truth; no live local model evaluation; no industrial vendor/model breadth study; no penetration test; no production load test; and no human time study. The test client emits a Starlette warning that the installed HTTPX integration is deprecated, though tests pass. Docker Compose could not be run because a daemon was unavailable.

Next evaluation protocol

Recruit OT engineers from multiple operators; pre-label a larger, source-reviewed corpus across vendors and versions; blind reviewers to system decisions; adjudicate disagreements; report precision, recall, false-negative rate, abstention coverage, calibration only if statistically supported, source traceability, and review minutes. Run the deterministic baseline first. Evaluate the LLM separately against gold spans and exact field values. Test stale/offline feeds, large inventory imports, model-output injection, RBAC, and deployment security.

The current result is a build check, not a customer study, security certification, safety case, or verified reduction in operational risk.